

Gianfranco Alarcon

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Education

University of Central Florida

Bachelor of Engineering in Computer Engineering; GPA: 3.3

Orlando, FL | August 2022 - December 2026

Relevant Coursework: Engineering Computation and Analysis, Data Structures and Algorithms, Operating Systems, Computer Architecture, Software Design, Machine Learning, IoT System Design

Experience

JLB Services, Manchester, United Kingdom

Freelance Development | March 2025 - Present

- Designed and built a custom delivery-service booking web app using React, Node.js and RESTful APIs
- Implemented backend logic and database integration (via Supabase) to store user orders, manage user sessions, and enable retrieval of past bookings

VR Orlando Rentals, Orlando, FL

Coding Teacher | Jan 2023 - May 2023

- Delivered programming/engineering lessons to K-12 students using Scratch and VR
- Collaborated with program leadership to coordinate classroom logistics and ensure day-to-day operations across multiple classrooms.

Projects

KnightHacks Competition: Project Artemis

Team Leader/Embedded Systems Developer | January 2025 - April 2025

- Implemented a communication link from Arduino to Raspberry Pi for real-time turret control
- Instantiated a web-server for camera vision functionality via Python
- Collaborated in a cross-functional team to align hardware-software communication and optimize responsiveness in object detection and servo actuation via C++

Sharedle Wordle Clone

Processes of Object Oriented Software Design - Backend Developer | August 2025 - November 2025

- Developed a full-stack multi-platform Wordle-style word guessing game with backend APIs for word validation, score tracking, and user authentication via MongoDB
- Designed RESTful endpoints and integrated MongoDB database with backend via JSON-based communication with Node.js

Language Learning App

Full-stack Developer | January 2025 - Present

- Developed frontend with React and backend with Node.js/MySQL
- Integrated ASR-LLM pipeline to extract audio and semantic meaning from Russian speech.

FreeRTOS EFR32xG24 Temperature/Humidity/Lux IoT Design

IoT System Design | January 2026 - Present

- Implementing an IoT device that reads data from temperature/humidity and lux sensors via I2C connections and displaying via UART.
- Using FreeRTOS and Bluetooth module installed capabilities to communicate with data via phone application in React Native

RISC-V FPGA Installation

Digital Systems, Basys3, Vivado | March 2024 - Present

- Implementing a custom FPGA-based RISC-V system using the PicoRV32 core, enabling low-level control through memory-mapped I/O and hardware-software integration.
- Developing embedded firmware using RISC-V GCC and linker scripts, interfacing directly with hardware peripherals (LEDs, UART) for real-time control and debugging

Skills

Skills: C/C++, JS/HTML/CSS, React, Typescript, MySQL, MongoDB, Supabase, Git, Linux, Windows, Python, Java, FreeRTOS, Arduino, MSP430, EFR32xG24, Raspberry Pi

Languages: English, Spanish, Russian